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BTech CSE First Year

PROJECT NAME:

MEDSAFE-AI-
AI-DRIVEN-MEDICAL-SAFETY-ASSISTANT.

GROUP MEMBERS:

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4. Sajid Kazi

AI-Based Medicine Safety Assistant

1. PROJECT OVERVIEW :

MedSafe AI is a healthcare safety assistant developed using Python and Streamlit. The system helps users analyze medicine interactions, extract medicines from prescriptions using OCR, understand symptoms, monitor side-effects, and estimate emergency health risks. The goal of the project is to provide educational health guidance while avoiding diagnostic claims.

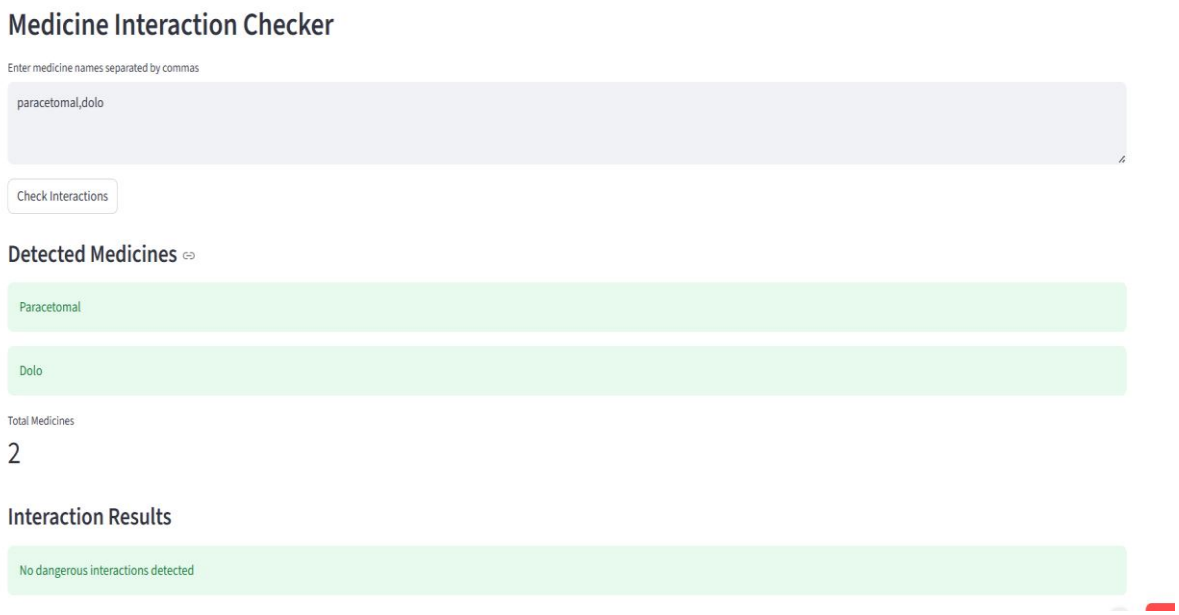
2. TECHNOLOGIES USED

- Python
- Streamlit (User Interface)
- OCR Processing (Prescription Text Extraction)
- Fuzzy Matching for Medicine Names
- AI/LLM for Explanation Generation
- JSON Data Handling
- Session State Management in Streamlit

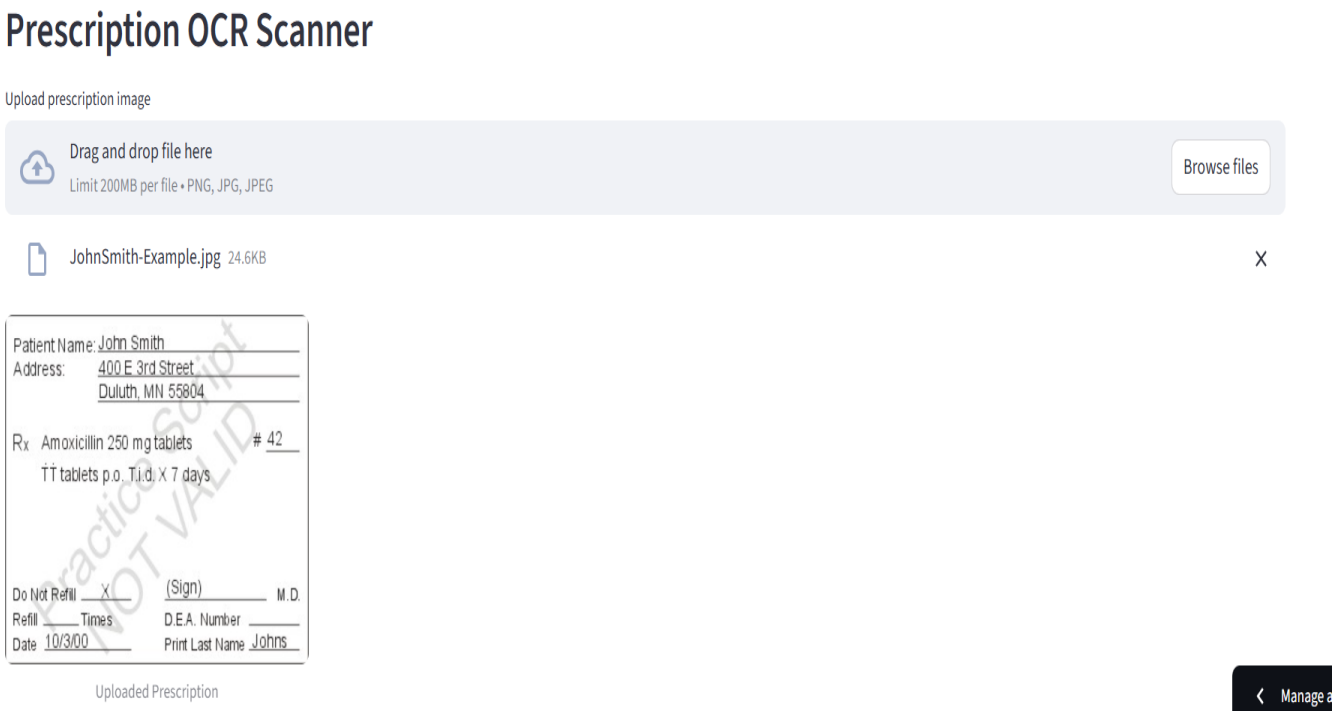
3. SYSTEM MODULES :

The application consists of five main modules:

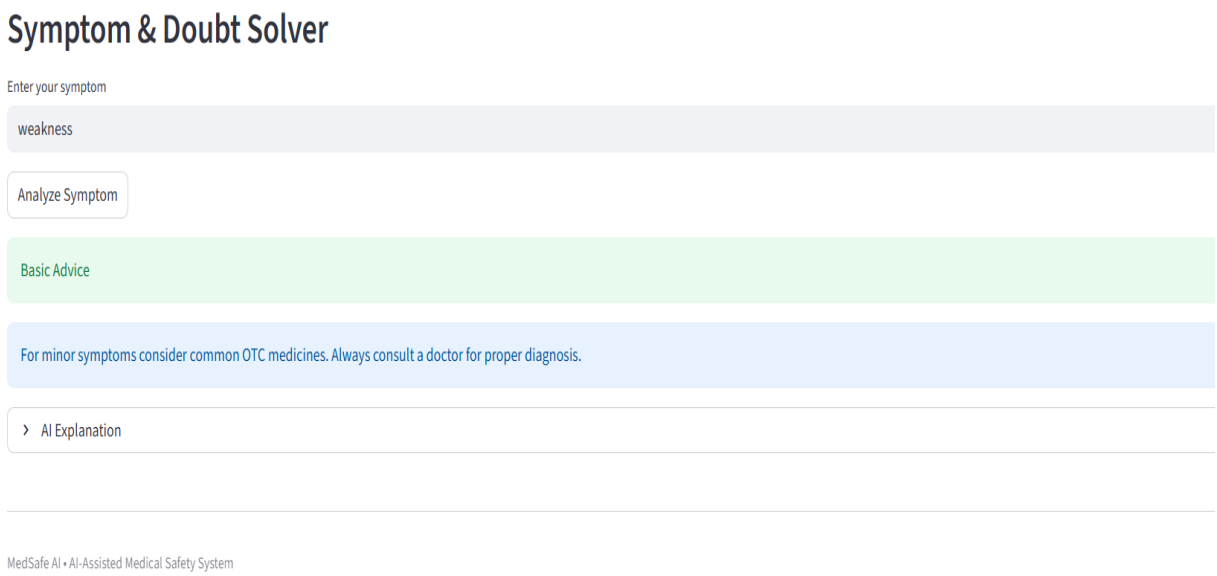
1. Medicine Interaction Checker – Detects possible interactions between medicines.



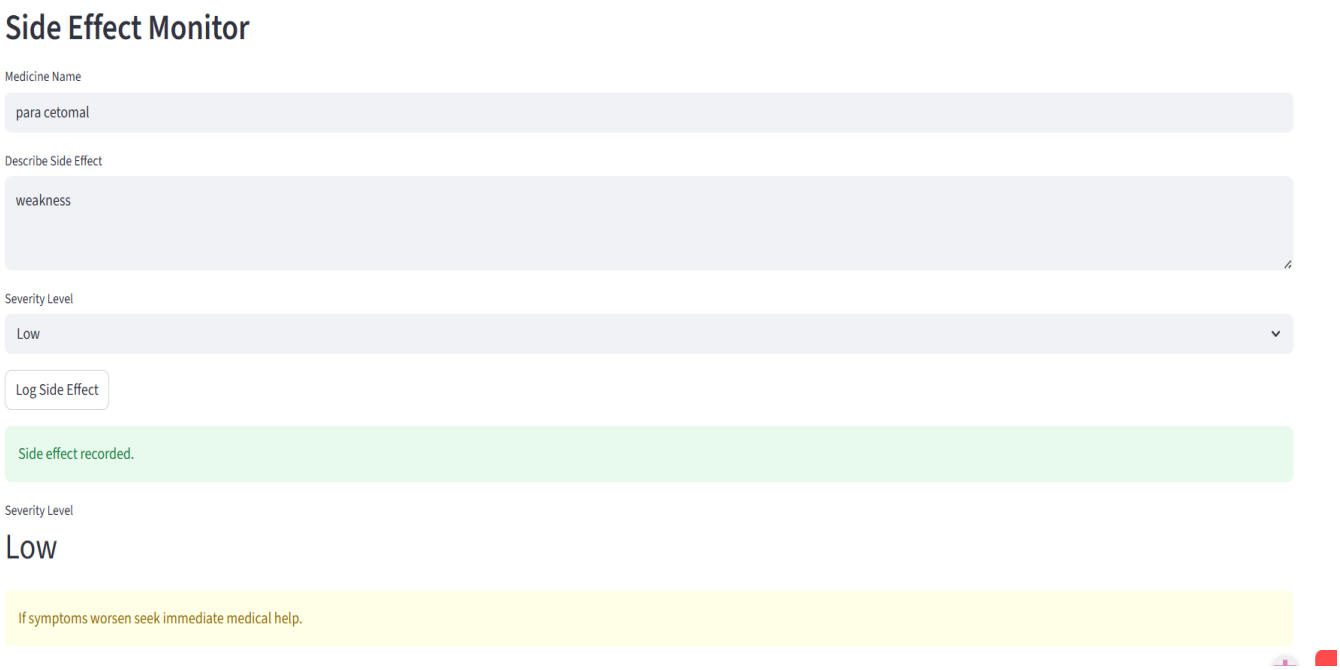
2. Prescription OCR – Extracts medicine names from uploaded prescription images:



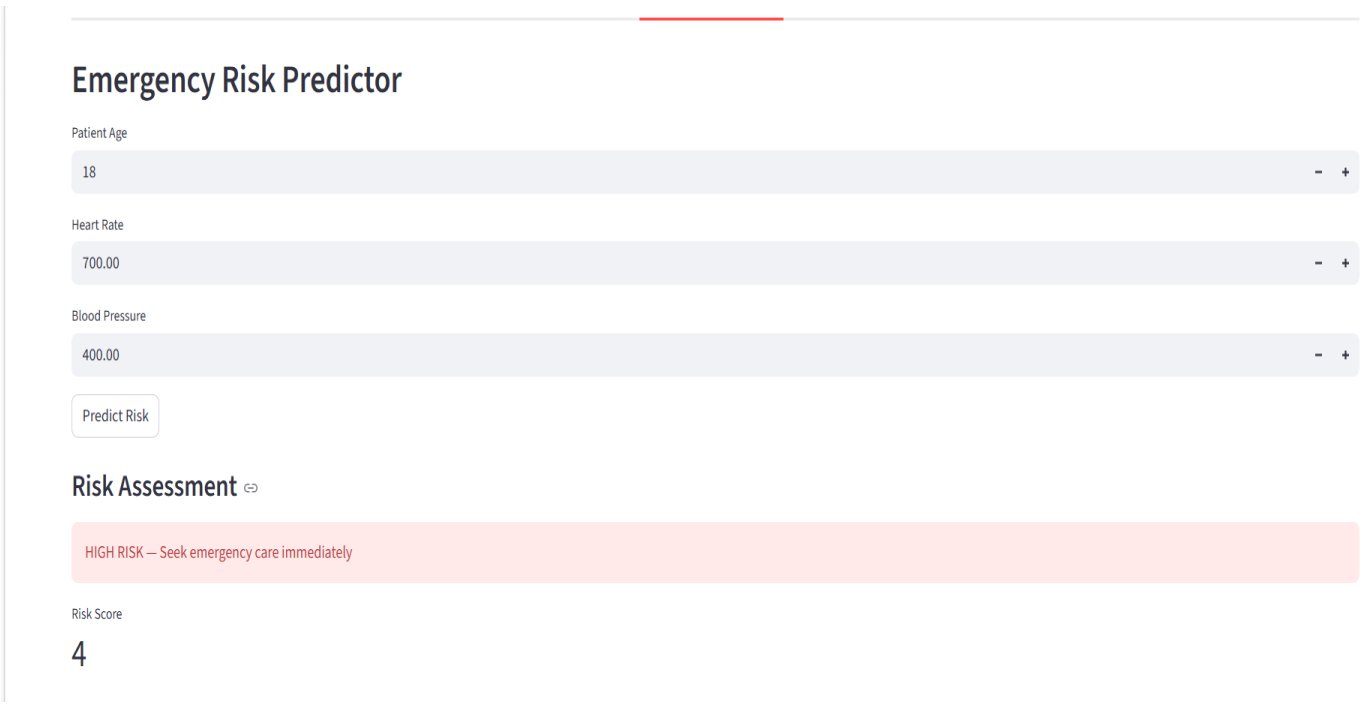
3.Symptom & Doubt Solver – Provides AI-based explanations for health questions.



4.Side-Effect Monitor – Allows users to log and review side-effect experiences.



5.Emergency Risk Predictor – Estimates emergency risk level based on symptoms and vital indicator.



4. USER INTERFACE DESIGN :

The interface is built using a structured multi-tab layout in Streamlit. Each tab represents a specific functionality of the system. The layout ensures clear separation of modules, readable typography, and responsive design for different screen sizes.

5. SUPPORTED DEPLOYMENT PLATFORMS:

The project can be published on the following platforms:

(Local Deployment)

Run the application on a local machine using the command:

(Copy code)

streamlit run streamlit_app.py

The system becomes accessible at:

(Copy code)

http://localhost:8501

Cloud Deployment:

The application can be published using Streamlit Community Cloud.

Source code is uploaded to GitHub.

The cloud platform automatically installs dependencies from requirements.txt.

6. TESTING AND VALIDATION

The application was tested for:

- Text input fields
- File upload functionality
- Button triggers• Tab navigation
- AI explanation generation
- Risk scoring logic
- Error handling behavior

7.Final Deployment Workflow

- The final publishing process follows these steps:
- Upload project code to GitHub repository
- Connect repository to Streamlit Cloud
- Configure dependencies using requirements.txt
- Set environment variables if required
- Deploy the application
- Share the generated public URL with users

8.Post-Deployment Validation:

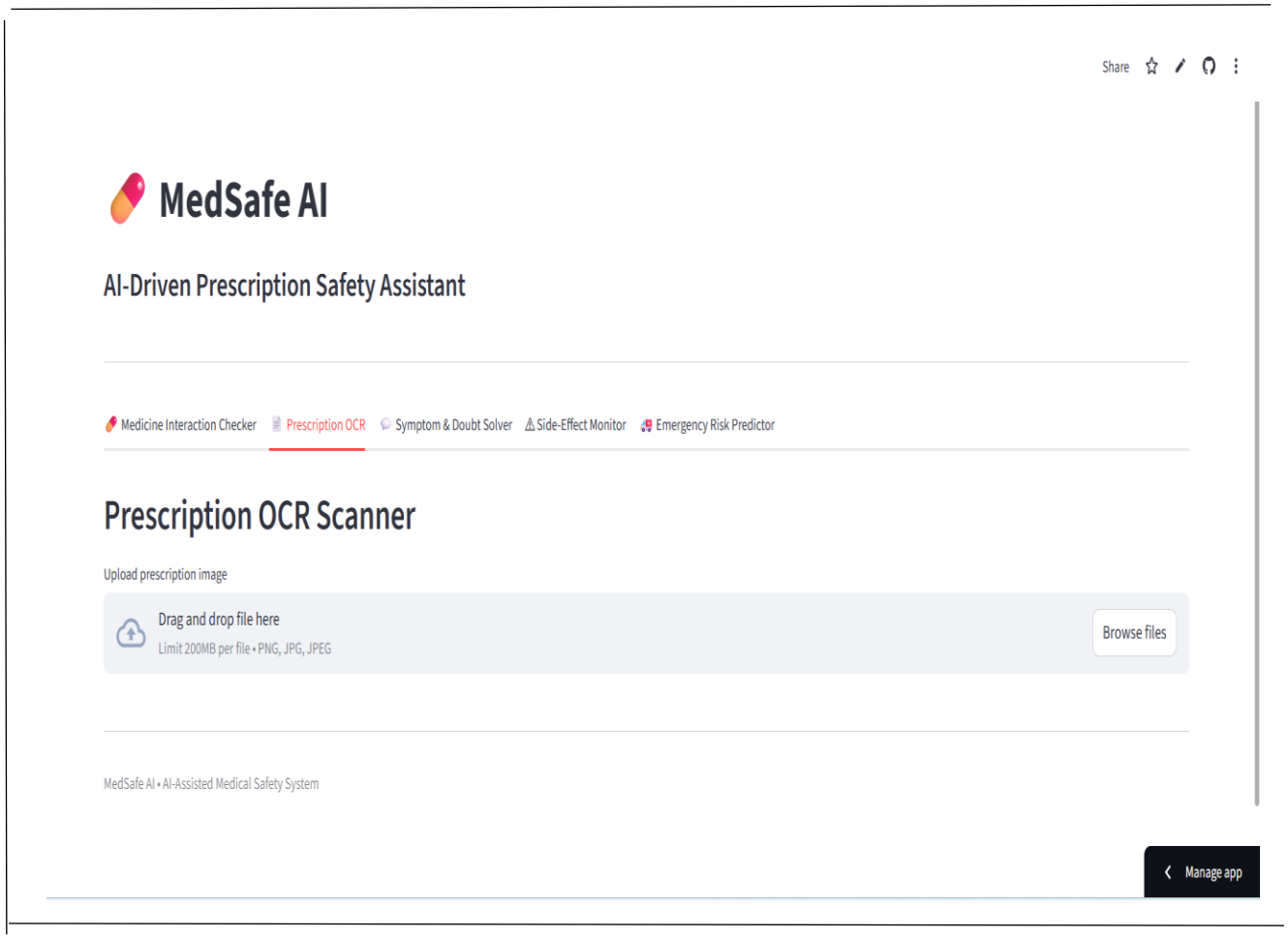
After publishing, the following checks are performed:

- Verify all modules load correctly
- Test OCR prescription extraction
- Validate medicine interaction detection
- Check AI explanation generation
- Ensure UI responsiveness across devices

9. FUTURE IMPROVEMENTS :

- Integration with real drug databases
- Improved OCR accuracy
- Mobile-friendly UI
- Better visualization of risk scores
- Secure storage of user health logs.

10. CONCLUSION

MedSafe AI demonstrates how AI tools can assist users in understanding medication safety and health risks. The system focuses on educational guidance, clear explanations, and safe user interaction while avoiding medical diagnosis.

❖ Final Project Layout: